

DMS613: Introduction to Mathematical Finance

Risk-Neutral Pricing

Sourav Majumdar

April 27, 2026

1 Assets, Portfolios, Arbitrage, and the Law of One Price

We begin with the simplest possible financial market: a one-period model. It is simple enough to write everything down explicitly, but rich enough to introduce three ideas that will stay with us throughout the course:

1. how to describe assets and portfolios,
2. what arbitrage means mathematically,
3. why equal payoffs should have equal prices.

1.1 A One-Period Financial Market

Consider a market with $d + 1$ traded assets. Trading takes place at two dates:

$$t = 0 \quad \text{and} \quad t = 1.$$

At time 0, the price of asset i is denoted by $\pi^i \geq 0$. We collect these prices in the vector

$$\bar{\pi} = (\pi^0, \pi^1, \dots, \pi^d) \in \mathbb{R}_+^{d+1}.$$

At time 1, the asset prices are uncertain. To model this uncertainty we work on a probability space

$$(\Omega, \mathcal{F}, P).$$

Each $\omega \in \Omega$ represents a possible state of the world. The time-1 payoff of asset i is a random variable $S^i(\omega)$, and we write

$$\bar{S} = (S^0, S^1, \dots, S^d).$$

So the market data are simply this: a deterministic price vector today and a random payoff vector tomorrow.

We assume that asset 0 is a risk-free bond satisfying

$$\pi^0 = 1, \quad S^0 = 1 + r,$$

for some interest rate $r > -1$.

In words: one unit invested in the bond at time 0 grows deterministically to $1 + r$ at time 1.

1.2 Portfolios

At time 0 an investor chooses a portfolio

$$\bar{\xi} = (\xi^0, \xi^1, \dots, \xi^d) \in \mathbb{R}^{d+1}.$$

Here ξ^i denotes the number of units of asset i held in the portfolio. Positive values mean long positions, while negative values correspond to short positions.

The cost of this portfolio at time 0 is

$$\bar{\pi} \cdot \bar{\xi} = \sum_{i=0}^d \pi^i \xi^i.$$

Its time-1 payoff in state ω is

$$\bar{\xi} \cdot \bar{S}(\omega) = \sum_{i=0}^d \xi^i S^i(\omega).$$

Using the special form of the bond, it is often convenient to write this as

$$\bar{\xi} \cdot \bar{S}(\omega) = \xi^0(1+r) + \xi \cdot S(\omega),$$

where $\xi = (\xi^1, \dots, \xi^d)$ and $S = (S^1, \dots, S^d)$ collect only the risky assets.

1.3 The Law of One Price

A very basic economic principle governs sensible markets.

Intuition. If two portfolios produce exactly the same payoff in every state of the world, then they should cost the same today. If they did not, an investor could buy the cheaper one, sell the more expensive one, and lock in a profit without bearing any risk.

Definition 1.1 (Law of One Price). *The market satisfies the law of one price if whenever two portfolios $\bar{\xi}$ and $\bar{\eta}$ satisfy*

$$\bar{\xi} \cdot \bar{S}(\omega) = \bar{\eta} \cdot \bar{S}(\omega) \quad \forall \omega \in \Omega,$$

then

$$\bar{\pi} \cdot \bar{\xi} = \bar{\pi} \cdot \bar{\eta}.$$

If two portfolios have identical payoffs, then their difference has zero payoff in every state. So the law of one price can be restated in a slightly more compact way.

Proposition 1.1. *The law of one price holds if and only if every portfolio with zero payoff in all states has zero price; that is,*

$$\bar{\xi} \cdot \bar{S}(\omega) = 0 \quad \forall \omega \in \Omega \quad \implies \quad \bar{\pi} \cdot \bar{\xi} = 0.$$

1.4 Arbitrage Opportunities

The law of one price is important, but by itself it is not yet the full story. What we really want to rule out is the possibility of making money for free. That leads to the notion of arbitrage.

Definition 1.2. A portfolio $\bar{\xi}$ is called an arbitrage opportunity if

$$\bar{\pi} \cdot \bar{\xi} \leq 0,$$

$$\bar{\xi} \cdot \bar{S}(\omega) \geq 0 \quad \forall \omega \in \Omega,$$

and

$$\mathbb{P}(\bar{\xi} \cdot \bar{S} > 0) > 0.$$

So an arbitrage opportunity has three features:

1. it does not require a positive initial investment,
2. it never loses money,
3. it makes a strictly positive profit with positive probability.

That is about as close to a free lunch as mathematics allows.

1.5 A Useful Characterization of Arbitrage

The next lemma rewrites the arbitrage condition in terms of the risky assets alone. It is the version that is often easier to check in applications.

Lemma 1.1. The following statements are equivalent.

1. The market admits an arbitrage opportunity.
2. There exists $\xi \in \mathbb{R}^d$ such that

$$\xi \cdot S \geq (1 + r) \xi \cdot \pi \quad \mathbb{P}\text{-a.s.},$$

and

$$\mathbb{P}(\xi \cdot S > (1 + r) \xi \cdot \pi) > 0.$$

Proof. We prove (1) $\Rightarrow$ (2) and (2) $\Rightarrow$ (1).

(1) $\Rightarrow$ (2). Suppose the market admits an arbitrage opportunity. Then there exists a portfolio

$$\bar{\xi} = (\xi^0, \xi) \in \mathbb{R}^{d+1}$$

such that

$$\bar{\pi} \cdot \bar{\xi} \leq 0, \quad \bar{\xi} \cdot \bar{S} \geq 0 \quad \mathbb{P}\text{-a.s.}, \quad \mathbb{P}(\bar{\xi} \cdot \bar{S} > 0) > 0.$$

Since

$$\bar{\pi} \cdot \bar{\xi} = \xi^0 + \xi \cdot \pi,$$

the condition $\bar{\pi} \cdot \bar{\xi} \leq 0$ implies

$$\xi^0 \leq -\xi \cdot \pi.$$

Also, because $S^0 = 1 + r$,

$$\bar{\xi} \cdot \bar{S} = \xi^0(1 + r) + \xi \cdot S.$$

Multiplying $\xi^0 \leq -\xi \cdot \pi$ by $1 + r > 0$ gives

$$(1 + r)\xi^0 \leq -(1 + r)\xi \cdot \pi.$$

Therefore,

$$\xi \cdot S - (1 + r)\xi \cdot \pi \geq \xi \cdot S + (1 + r)\xi^0 = \bar{\xi} \cdot \bar{S}.$$

Since $\bar{\xi} \cdot \bar{S} \geq 0$ almost surely, we obtain

$$\xi \cdot S \geq (1 + r)\xi \cdot \pi \quad \mathbb{P}\text{-a.s.}$$

Moreover,

$$\xi \cdot S - (1 + r)\xi \cdot \pi \geq \bar{\xi} \cdot \bar{S},$$

so from $\mathbb{P}(\bar{\xi} \cdot \bar{S} > 0) > 0$ we also get

$$\mathbb{P}(\xi \cdot S > (1 + r)\xi \cdot \pi) > 0.$$

Thus condition (2) holds.

(2) $\Rightarrow$ (1). Now suppose there exists $\xi \in \mathbb{R}^d$ such that

$$\xi \cdot S \geq (1 + r)\xi \cdot \pi \quad \mathbb{P}\text{-a.s.},$$

and

$$\mathbb{P}(\xi \cdot S > (1 + r)\xi \cdot \pi) > 0.$$

Define a portfolio $\bar{\xi} = (\xi^0, \xi)$ by choosing the bond holding as

$$\xi^0 := -\xi \cdot \pi.$$

Then its initial cost is

$$\bar{\pi} \cdot \bar{\xi} = \xi^0 + \xi \cdot \pi = 0.$$

Its terminal payoff is

$$\bar{\xi} \cdot \bar{S} = \xi^0(1 + r) + \xi \cdot S = -(1 + r)\xi \cdot \pi + \xi \cdot S.$$

By assumption this is nonnegative almost surely, and strictly positive with positive probability. Hence $\bar{\xi}$ is an arbitrage opportunity. $\square$

1.6 Examples

A few examples help build intuition.

Example 1

Suppose there are two assets with prices

$$\pi = (1, 2)$$

and payoffs

$$S(\omega_1) = (2, 1), \quad S(\omega_2) = (0, 1).$$

Consider the portfolio $\xi = (-1, 1)$. Its initial cost is

$$\pi \cdot \xi = -1 + 2 = 1.$$

Its payoffs are

$$\xi \cdot S(\omega_1) = -2 + 1 = -1, \quad \xi \cdot S(\omega_2) = 0 + 1 = 1.$$

So this portfolio can make money in one state and lose money in another. In particular, it is *not* an arbitrage opportunity. The point of the example is simply that not every apparently clever trade is free money; some are just ordinary risky bets.

Example 2

Consider prices

$$\pi = (1, 2)$$

and payoff matrix

$$\begin{pmatrix} -2 & -4 \\ -1 & -2 \end{pmatrix}.$$

If we short asset 1, that is, choose

$$\xi = (-1, 0),$$

then the initial cost is

$$\pi \cdot \xi = -1,$$

so we receive one unit of cash today. The terminal payoffs are

$$\xi \cdot S(\omega_1) = 2, \quad \xi \cdot S(\omega_2) = 4.$$

Thus we receive money at time 0 and also end up with a nonnegative, in fact strictly positive, payoff in every state. This is an arbitrage opportunity.

Example 3

Suppose an asset has price 0 today but delivers payoff 1 in some state. Then buying one unit of this asset costs nothing and gives a positive payoff with positive probability. That is immediate arbitrage.

Example 4

Consider three assets with prices

$$\pi = (1, 1, 1)$$

and state-by-state payoffs

$$S(\omega_1) = (1, 0, 0), \quad S(\omega_2) = (0, 1, 0), \quad S(\omega_3) = (0, 0, 1).$$

These are just state securities, each priced at 1. Any nonnegative payoff vector is obtained by taking nonnegative holdings of these securities, and the cost is exactly the sum of those holdings. So there is no way to build a portfolio with nonnegative payoffs in all states and nonpositive cost unless all holdings are zero. Hence there is no arbitrage in this market.

Example 5

Let

$$\pi = (1, 2, 3)$$

and

$$S(\omega_1) = (1, 2, 3), \quad S(\omega_2) = (2, 4, 6).$$

Here asset 2 is exactly twice asset 1, and asset 3 is exactly three times asset 1, both in every state. The law of one price therefore requires

$$\pi_2 = 2\pi_1, \quad \pi_3 = 3\pi_1.$$

Since these identities hold, the market is consistent with the law of one price. If, say, asset 2 were priced at something other than $2\pi_1$, then we could exploit the mismatch by buying the cheaper payoff and selling the more expensive one.

These examples are deliberately simple, but they already show the general pattern: arbitrage appears when prices are inconsistent with payoffs.

2 Risk-Neutral pricing in a discrete model

We now shift perspective. In the previous section we focused on portfolios, replication, and arbitrage in a one-period market. We now introduce a second, and extremely powerful, way of thinking about prices: *risk-neutral pricing*.

The big idea is this. In an arbitrage-free market, it is often possible to find a probability measure under which discounted asset prices behave like martingales. Once that measure is available, derivative pricing becomes an expectation problem.

That sounds abstract at first, so we begin again in the one-period setting, where everything can be seen explicitly.

Recall the one-period market with initial price vector

$$\bar{\pi} = (\pi^0, \pi^1, \dots, \pi^d)$$

and random time-1 payoff vector

$$\bar{S} = (S^0, S^1, \dots, S^d).$$

We continue to assume that asset 0 is the bond, so

$$\pi^0 = 1, \quad S^0 = 1 + r,$$

with $r > -1$.

The absence of arbitrage does not just tell us that “free money” is impossible. It also imposes a very rigid structure on prices. In finite-state models this structure can be written in terms of *state prices*, and that is the cleanest route to the risk-neutral measure.

2.1 State Prices in a Finite One-Period Model

Suppose the state space is finite:

$$\Omega = \{\omega_1, \dots, \omega_n\}.$$

A *state security* for state ω_j is a hypothetical asset that pays

$$1 \quad \text{in state } \omega_j, \quad 0 \quad \text{in all other states.}$$

Let q_j be its time-0 price. The numbers $q_1, \dots, q_n$ are called the *state prices*.

Now suppose a claim pays $X(\omega_j)$ in state ω_j . If state securities are available, then we can replicate this claim by buying $X(\omega_j)$ units of the j th state security. So its price must be

$$\pi(X) = \sum_{j=1}^n q_j X(\omega_j).$$

In particular, for asset i we obtain

$$\pi^i = \sum_{j=1}^n q_j S^i(\omega_j), \quad i = 0, 1, \dots, d.$$

Apply this to the bond. Since $S^0(\omega_j) = 1 + r$ in every state,

$$1 = \pi^0 = \sum_{j=1}^n q_j (1 + r) = (1 + r) \sum_{j=1}^n q_j.$$

Hence

$$\sum_{j=1}^n q_j = \frac{1}{1 + r}.$$

This normalization suggests defining

$$\mathbb{P}^*(\omega_j) := (1+r)q_j, \quad j = 1, \dots, n.$$

Then

$$\mathbb{P}^*(\omega_j) \geq 0, \quad \sum_{j=1}^n \mathbb{P}^*(\omega_j) = 1,$$

so $\mathbb{P}^*$ is a probability measure on Ω .

Substituting $q_j = \mathbb{P}^*(\omega_j)/(1+r)$ into the pricing formula gives

$$\pi^i = \frac{1}{1+r} \sum_{j=1}^n \mathbb{P}^*(\omega_j) S^i(\omega_j) = \frac{1}{1+r} E^*[S^i],$$

where E^* denotes expectation under $\mathbb{P}^*$.

Definition 2.1 (Risk-Neutral Measure in the One-Period Model). *A probability measure $\mathbb{P}^*$ on $(\Omega, \mathcal{F})$ is called a risk-neutral measure if*

$$\pi^i = E^* \left[\frac{S^i}{1+r} \right], \quad i = 0, 1, \dots, d.$$

Interpretation. Under $\mathbb{P}^*$, the time-0 price of every asset is the expectation of its discounted payoff. This is why the measure is called *risk-neutral*: once we discount by the bond, assets have no expected drift under this pricing measure.

A small but important warning. $\mathbb{P}^*$ need not be the “true” probability measure describing how the world actually evolves. It is a *pricing measure*, not necessarily a physical or statistical one.

2.2 Discounted Prices and the Martingale Viewpoint

In the one-period setting, define the discounted payoff

$$X^i := \frac{S^i}{1+r}.$$

Then the pricing relation reads

$$\pi^i = E^*[X^i].$$

In continuous time, the corresponding statement becomes a conditional expectation identity: under a suitable measure Q ,

$$E^Q[\tilde{S}(t) \mid \mathcal{F}(s)] = \tilde{S}(s), \quad 0 \leq s \leq t,$$

where $\tilde{S}(t)$ denotes the discounted asset price. In other words, *discounted prices are martingales under the pricing measure*.

This is one of the central mantras of the course.

2.3 The One-Period Binomial Model

Let us now see the risk-neutral idea in the simplest possible example. Suppose there is one risky asset (a stock) and one risk-free asset (a bond). At time $t = 0$ the stock price is $S(0)$, and at time $t = 1$ the stock can move to one of two values:

$$S(1) = \begin{cases} uS(0), & \text{in the up state,} \\ dS(0), & \text{in the down state,} \end{cases}$$

where

$$u > d > 0.$$

The bond satisfies

$$B(0) = 1, \quad B(1) = 1 + r,$$

with $r > -1$.

When is the model arbitrage-free?

A necessary and sufficient condition for absence of arbitrage is

$$d < 1 + r < u.$$

This condition says that the bond return must lie strictly between the stock's down return and up return.

Let us check this directly.

A portfolio consists of Δ units of stock and β units of the bond. Thus

$$\bar{\xi} = (\beta, \Delta).$$

Its initial cost is

$$\beta + \Delta S(0),$$

and its time-1 payoff is

$$\begin{cases} \beta(1+r) + \Delta u S(0), & \text{up state,} \\ \beta(1+r) + \Delta d S(0), & \text{down state.} \end{cases}$$

Case 1: $u \leq 1 + r$. Take

$$\Delta = -1, \quad \beta = S(0).$$

So we short one share and invest the proceeds in the bond. The initial cost is zero, and the time-1 payoffs are

$$S(0)((1+r) - u) \quad \text{and} \quad S(0)((1+r) - d).$$

These are both nonnegative, and the second is strictly positive because $d < u$. Hence this is an arbitrage opportunity.

Case 2: $d \geq 1 + r$. Take

$$\Delta = 1, \quad \beta = -S(0).$$

So we borrow to buy one share of stock. Again the initial cost is zero, and the time-1 payoffs are

$$S(0)(u - (1+r)) \quad \text{and} \quad S(0)(d - (1+r)).$$

These are both nonnegative, and the first is strictly positive because $u > d$. So this is also an arbitrage opportunity.

Case 3: $d < 1 + r < u$. We now show that no arbitrage can exist.

Suppose (β, Δ) is an arbitrage opportunity. Then

$$\beta + \Delta S(0) \leq 0,$$

$$\beta(1 + r) + \Delta uS(0) \geq 0, \quad \beta(1 + r) + \Delta dS(0) \geq 0,$$

with at least one strict inequality.

If $\Delta > 0$, then from $\beta + \Delta S(0) \leq 0$ we get

$$\beta \leq -\Delta S(0).$$

Multiplying by $1 + r$ and using the down-state payoff,

$$\beta(1 + r) + \Delta dS(0) \leq \Delta S(0)(d - (1 + r)) < 0,$$

a contradiction.

If $\Delta < 0$, then similarly the up-state payoff satisfies

$$\beta(1 + r) + \Delta uS(0) \leq \Delta S(0)(u - (1 + r)) < 0,$$

again a contradiction.

Finally, if $\Delta = 0$, then $\beta \leq 0$, so the payoff in each state is $\beta(1 + r) \leq 0$. For it to be nonnegative in every state we must have $\beta = 0$, but then the payoff is identically zero, so we do not have arbitrage.

Thus the one-period binomial model is arbitrage-free if and only if

$$d < 1 + r < u.$$

Computation of the Risk-Neutral Probability

Now we look for a probability p^* such that the current stock price equals the discounted expected future stock price:

$$S(0) = \frac{1}{1 + r} E^*[S(1)].$$

Since $S(1)$ takes the values $uS(0)$ and $dS(0)$, this becomes

$$S(0) = \frac{1}{1 + r} (p^* uS(0) + (1 - p^*) dS(0)).$$

Dividing through by $S(0) > 0$ yields

$$1 + r = p^* u + (1 - p^*) d.$$

Solving for p^* gives

$$p^* = \frac{(1 + r) - d}{u - d}.$$

This is the *risk-neutral probability*.

Why is this really a probability?

Because $d < 1 + r < u$, we have

$$0 < (1 + r) - d < u - d,$$

so

$$0 < p^* < 1.$$

Thus p^* is indeed a genuine probability.

Discounted Stock Price as a Martingale

Define the discounted stock price by

$$\tilde{S}(t) = \frac{S(t)}{B(t)}.$$

Since $B(0) = 1$ and $B(1) = 1 + r$,

$$\tilde{S}(0) = S(0), \quad \tilde{S}(1) = \frac{S(1)}{1 + r}.$$

Under $\mathbb{P}^*$,

$$E^*[\tilde{S}(1)] = \frac{1}{1 + r} (p^* u S(0) + (1 - p^*) d S(0)) = S(0) = \tilde{S}(0).$$

So the discounted stock price is a martingale under the risk-neutral measure.

Pricing a Derivative in the Binomial Model

Let a derivative have payoff

$$V(1) = \begin{cases} V_u, & \text{up state,} \\ V_d, & \text{down state.} \end{cases}$$

Then its no-arbitrage price at time 0 is

$$V(0) = \frac{1}{1 + r} (p^* V_u + (1 - p^*) V_d).$$

So the derivative price is the discounted expectation of the payoff under the risk-neutral probability.

This is exactly the finite-state prototype of the continuous-time pricing formula

$$V(t) = E^Q \left[e^{-r(T-t)} V(T) \mid \mathcal{F}(t) \right].$$

Remark 1. *The binomial model makes two ideas completely transparent:*

1. *the risk-neutral probability is determined by no-arbitrage, not by actual empirical distribution.*
2. *under that probability, the discounted stock price is a martingale.*

These two principles survive unchanged in continuous time.

3 Continuous-time pricing

Up to this point, everything we have done has a very clear structure. In the one-period model, absence of arbitrage led us to the existence of a risk-neutral probability measure $\mathbb{P}^*$ under which prices satisfy

$$\pi = \frac{1}{1 + r} E^*[S].$$

There are two key takeaways from the discrete theory:

1. Pricing is not driven by the physical probability measure P . The real-world drift of the asset does not enter the pricing formula.
2. Discounted prices must behave like martingales. That is, under the correct pricing measure, they must be martingales.

When we move to continuous time, we would like to preserve exactly these two principles. However, a difficulty immediately appears.

Recall that under the physical measure P , the stock price is modeled by the geometric Brownian motion

$$dS(t) = \mu S(t) dt + \sigma S(t) dW(t), \quad \sigma > 0,$$

and the money market account is

$$B(t) = e^{rt}.$$

As before, define the discounted stock price by

$$\tilde{S}(t) := \frac{S(t)}{B(t)} = e^{-rt} S(t).$$

A direct application of Itô's formula gives

$$d\tilde{S}(t) = (\mu - r)\tilde{S}(t) dt + \sigma\tilde{S}(t) dW(t).$$

So under the original probability measure P , the discounted stock price still has a drift term unless $\mu = r$. In particular, $\tilde{S}(t)$ is generally not a martingale under P .

But from the discrete-time theory, we know what we would like to have in an arbitrage-free pricing framework, that under the correct pricing measure, discounted asset prices should behave like martingales. This suggests that the problem is not with the process $S(t)$ itself, but with the probability measure under which we are viewing it.

The idea is therefore to modify the Brownian motion in such a way that the drift term disappears. Write

$$\theta := \frac{\mu - r}{\sigma}.$$

Now define a new process by

$$W^Q(t) := W(t) + \theta t.$$

Then formally,

$$dW(t) = dW^Q(t) - \theta dt.$$

Substituting this into the dynamics of $\tilde{S}(t)$, we obtain

$$d\tilde{S}(t) = (\mu - r)\tilde{S}(t) dt + \sigma\tilde{S}(t)(dW^Q(t) - \theta dt).$$

Since $\sigma\theta = \mu - r$, the drift terms cancel:

$$d\tilde{S}(t) = \sigma\tilde{S}(t) dW^Q(t).$$

So at a purely formal level, we have found exactly the correction needed to remove the drift. Equivalently, the original stock-price dynamics become

$$dS(t) = rS(t) dt + \sigma S(t) dW^Q(t).$$

This is the risk-neutral form we were hoping for.

However, an important question still remains, i.e., why is the process W^Q actually a Brownian motion? After all, we have simply written down a drift-corrected process. For risk-neutral pricing, it is not enough to cancel the drift algebraically. We must also know that under a suitable new probability measure, the process W^Q really is Brownian motion. This is where the Girsanov's theorem enters.

3.1 Girsanov's theorem

Theorem 3.1 (Girsanov's theorem). *Let $(W(t))_{0 \leq t \leq T}$ be a Brownian motion on $(\Omega, \mathcal{F}, (\mathcal{F}(t))_{0 \leq t \leq T}, P)$, and let $\Theta(t)$ be an adapted process satisfying*

$$E \left[\exp \left(\frac{1}{2} \int_0^T \Theta(u)^2 du \right) \right] < \infty.$$

Define

$$Z(t) = \exp \left\{ - \int_0^t \Theta(u) dW(u) - \frac{1}{2} \int_0^t \Theta(u)^2 du \right\}, \quad 0 \leq t \leq T.$$

Then $(Z(t))_{0 \leq t \leq T}$ is a positive martingale and

$$E[Z(T)] = 1.$$

Hence we may define a new probability measure Q on $\mathcal{F}(T)$ by

$$Q(A) = E[1_A Z(T)], \quad A \in \mathcal{F}(T).$$

Under this measure Q , the process

$$W^Q(t) := W(t) + \int_0^t \Theta(u) du$$

is a Brownian motion.

The random variable $Z(T)$ plays the role of a likelihood ratio between the two measures. More precisely,

$$\frac{dQ}{dP} = Z(T),$$

so that Q is absolutely continuous with respect to P , written $Q \ll P$. That is, for every event $A \in \mathcal{F}$,

$$P(A) = 0 \implies Q(A) = 0.$$

The theorem tells us that, if we choose the process Θ appropriately and define a new measure Q using the likelihood ratio process $Z(T)$, then the modified process

$$W^Q(t) = W(t) + \int_0^t \Theta(u) du$$

really is Brownian motion under Q . This is the rigorous justification for the risk-neutral change of measure.

Remark 2. *Girsanov's theorem is the continuous-time analogue of changing probabilities in the binomial model. Instead of adjusting a finite set of probabilities, we reweight entire paths of the process.*

3.2 Application to geometric Brownian motion

We now check that Girsanov's theorem applies to the geometric Brownian motion model above. In this case, the correction term is constant:

$$\Theta(t) \equiv \theta = \frac{\mu - r}{\sigma}.$$

Therefore

$$\int_0^T \Theta(u)^2 du = \theta^2 T,$$

and Novikov's condition becomes

$$E \left[\exp \left(\frac{1}{2} \int_0^T \Theta(u)^2 du \right) \right] = E \left[\exp \left(\frac{1}{2} \theta^2 T \right) \right] = \exp \left(\frac{1}{2} \theta^2 T \right) < \infty.$$

So the condition is automatically satisfied.

The density process is

$$Z(t) = \exp \left\{ -\theta W(t) - \frac{1}{2} \theta^2 t \right\}.$$

Since θ is constant, Girsanov's theorem gives a new probability measure Q such that

$$W^Q(t) = W(t) + \theta t$$

is a Brownian motion under Q .

3.3 Risk-neutral pricing formula

We now make precise the pricing principle suggested by the martingale viewpoint. Suppose a contingent claim matures at time T and has payoff $V(T)$. If the claim can be replicated by a self-financing trading strategy, then its price process must be obtained by discounting and taking conditional expectation under the risk-neutral measure.

Theorem 3.2 (Risk-neutral pricing formula). *Suppose a contingent claim with maturity T is replicated by a self-financing strategy $(\beta(u), \Delta(u))_{0 \leq u \leq T}$, where $\beta(u)$ denotes the number of units held in the bond and $\Delta(u)$ denotes the number of units held in the stock. Let $V(u)$ be the value process of this strategy:*

$$V(u) = \beta(u)B(u) + \Delta(u)S(u).$$

Assume moreover that the stochastic integral appearing below is a true martingale. Then for every $0 \leq t \leq T$,

$$V(t) = E^Q \left[e^{-r(T-t)} V(T) \mid \mathcal{F}(t) \right].$$

Proof. Since the strategy is self-financing,

$$dV(u) = \beta(u) dB(u) + \Delta(u) dS(u).$$

Under the risk-neutral measure Q , the bond and stock satisfy

$$dB(u) = rB(u) du, \quad dS(u) = rS(u) du + \sigma S(u) dW^Q(u).$$

Therefore

$$\begin{aligned} dV(u) &= \beta(u)rB(u) du + \Delta(u)(rS(u) du + \sigma S(u) dW^Q(u)) \\ &= r(\beta(u)B(u) + \Delta(u)S(u)) du + \sigma \Delta(u)S(u) dW^Q(u) \\ &= rV(u) du + \sigma \Delta(u)S(u) dW^Q(u). \end{aligned}$$

Now discount by the money market account. Using Itô's formula,

$$\begin{aligned} d(e^{-ru} V(u)) &= -re^{-ru} V(u) du + e^{-ru} dV(u) \\ &= -re^{-ru} V(u) du + e^{-ru} (rV(u) du + \sigma \Delta(u)S(u) dW^Q(u)) \\ &= e^{-ru} \sigma \Delta(u)S(u) dW^Q(u). \end{aligned}$$

Hence the discounted value process

$$\tilde{V}(u) := e^{-ru} V(u)$$

is a Q -martingale. Therefore,

$$\tilde{V}(t) = E^Q[\tilde{V}(T) \mid \mathcal{F}(t)].$$

That is,

$$e^{-rt} V(t) = E^Q[e^{-rT} V(T) \mid \mathcal{F}(t)].$$

Multiplying both sides by e^{rt} gives

$$V(t) = E^Q[e^{-r(T-t)} V(T) \mid \mathcal{F}(t)].$$

□

3.4 The Black–Scholes formula via risk-neutral pricing

We now apply the risk-neutral pricing formula to a European call option with strike K and maturity T . Its payoff is

$$C(T) = (S(T) - K)^+.$$

Theorem 3.3 (Black–Scholes formula). *Let $C(t)$ denote the time- t price of the European call option with payoff*

$$C(T) = (S(T) - K)^+.$$

Then

$$C(t) = S(t)N(d_+(t)) - Ke^{-r(T-t)}N(d_-(t)),$$

where

$$d_{\pm}(t) = \frac{\log\left(\frac{S(t)}{K}\right) + (r \pm \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}},$$

and $N(\cdot)$ denotes the standard normal distribution function,

$$N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-y^2/2} dy.$$

Proof. Set

$$\tau := T - t.$$

By the risk-neutral pricing formula,

$$C(t) = E^Q[e^{-r\tau}(S(T) - K)^+ \mid \mathcal{F}(t)].$$

So the task is to compute the conditional expectation explicitly.

Under Q , the stock price satisfies

$$dS(u) = rS(u) du + \sigma S(u) dW^Q(u), \quad t \leq u \leq T.$$

Apply Itô's formula to the function $f(x) = \log x$. Since

$$f'(x) = \frac{1}{x}, \quad f''(x) = -\frac{1}{x^2},$$

we obtain

$$d(\log S(u)) = \frac{1}{S(u)} dS(u) - \frac{1}{2} \frac{1}{S(u)^2} d[S, S](u).$$

Now

$$d[S, S](u) = \sigma^2 S(u)^2 du,$$

so substituting the dynamics of S gives

$$\begin{aligned} d(\log S(u)) &= \frac{1}{S(u)} (rS(u) du + \sigma S(u) dW^Q(u)) - \frac{1}{2} \frac{1}{S(u)^2} \sigma^2 S(u)^2 du \\ &= r du + \sigma dW^Q(u) - \frac{1}{2} \sigma^2 du \\ &= \left(r - \frac{1}{2} \sigma^2 \right) du + \sigma dW^Q(u). \end{aligned}$$

Integrating from t to T ,

$$\log S(T) - \log S(t) = \left(r - \frac{1}{2} \sigma^2 \right) \tau + \sigma (W^Q(T) - W^Q(t)).$$

Hence

$$\log S(T) = \log S(t) + \left(r - \frac{1}{2} \sigma^2 \right) \tau + \sigma (W^Q(T) - W^Q(t)),$$

and therefore

$$S(T) = S(t) \exp \left\{ \left(r - \frac{1}{2} \sigma^2 \right) \tau + \sigma (W^Q(T) - W^Q(t)) \right\}.$$

Because W^Q is Brownian motion under Q , the increment

$$W^Q(T) - W^Q(t)$$

is independent of $\mathcal{F}(t)$ and has distribution $N(0, \tau)$. Thus we may write

$$W^Q(T) - W^Q(t) = \sqrt{\tau} Z, \quad Z \sim N(0, 1),$$

where Z is independent of $\mathcal{F}(t)$. So

$$S(T) = S(t) \exp \left\{ \left(r - \frac{1}{2} \sigma^2 \right) \tau + \sigma \sqrt{\tau} Z \right\}.$$

Now split the payoff:

$$(S(T) - K)^+ = S(T) \mathbf{1}_{\{S(T) > K\}} - K \mathbf{1}_{\{S(T) > K\}}.$$

Hence

$$C(t) = e^{-r\tau} E^Q \left[S(T) \mathbf{1}_{\{S(T) > K\}} \mid \mathcal{F}(t) \right] - K e^{-r\tau} Q(S(T) > K \mid \mathcal{F}(t)).$$

We first compute the event $\{S(T) > K\}$. Since $S(T) > 0$,

$$S(T) > K \iff \log \left(\frac{S(T)}{S(t)} \right) > \log \left(\frac{K}{S(t)} \right).$$

Substituting the representation above,

$$\left(r - \frac{1}{2} \sigma^2 \right) \tau + \sigma \sqrt{\tau} Z > \log \left(\frac{K}{S(t)} \right).$$

Equivalently,

$$Z > \frac{\log\left(\frac{K}{S(t)}\right) - \left(r - \frac{1}{2}\sigma^2\right)\tau}{\sigma\sqrt{\tau}} = -d_-(t).$$

Therefore

$$Q(S(T) > K \mid \mathcal{F}(t)) = Q(Z > -d_-(t)) = N(d_-(t)).$$

Next we compute the first expectation:

$$E^Q\left[S(T)\mathbf{1}_{\{S(T) > K\}} \mid \mathcal{F}(t)\right] = S(t)e^{(r - \frac{1}{2}\sigma^2)\tau} E\left[e^{\sigma\sqrt{\tau}Z}\mathbf{1}_{\{Z > -d_-(t)\}}\right].$$

So it remains to evaluate

$$E\left[e^{aZ}\mathbf{1}_{\{Z > b\}}\right]$$

for a standard normal Z . Writing

$$\phi(z) = \frac{1}{\sqrt{2\pi}}e^{-z^2/2},$$

we have

$$\begin{aligned} E\left[e^{aZ}\mathbf{1}_{\{Z > b\}}\right] &= \int_b^\infty e^{az}\phi(z) dz \\ &= \frac{1}{\sqrt{2\pi}} \int_b^\infty \exp\left(az - \frac{1}{2}z^2\right) dz \\ &= \frac{1}{\sqrt{2\pi}} \int_b^\infty \exp\left(-\frac{1}{2}(z-a)^2 + \frac{1}{2}a^2\right) dz \\ &= e^{a^2/2} \frac{1}{\sqrt{2\pi}} \int_b^\infty e^{-(z-a)^2/2} dz. \end{aligned}$$

Now make the change of variable

$$y = z - a.$$

Then

$$\begin{aligned} E\left[e^{aZ}\mathbf{1}_{\{Z > b\}}\right] &= e^{a^2/2} \frac{1}{\sqrt{2\pi}} \int_{b-a}^\infty e^{-y^2/2} dy \\ &= e^{a^2/2} N(a - b). \end{aligned}$$

Apply this with

$$a = \sigma\sqrt{\tau}, \quad b = -d_-(t).$$

Then

$$\begin{aligned} E\left[e^{\sigma\sqrt{\tau}Z}\mathbf{1}_{\{Z > -d_-(t)\}}\right] &= e^{\sigma^2\tau/2} N(\sigma\sqrt{\tau} + d_-(t)) \\ &= e^{\sigma^2\tau/2} N(d_+(t)), \end{aligned}$$

because

$$d_+(t) = d_-(t) + \sigma\sqrt{\tau}.$$

Hence

$$\begin{aligned} E^Q\left[S(T)\mathbf{1}_{\{S(T) > K\}} \mid \mathcal{F}(t)\right] &= S(t)e^{(r - \frac{1}{2}\sigma^2)\tau} e^{\sigma^2\tau/2} N(d_+(t)) \\ &= S(t)e^{r\tau} N(d_+(t)). \end{aligned}$$

Substituting both pieces into the pricing formula,

$$\begin{aligned} C(t) &= e^{-r\tau} (S(t)e^{r\tau} N(d_+(t))) - Ke^{-r\tau} N(d_-(t)) \\ &= S(t)N(d_+(t)) - Ke^{-r(T-t)} N(d_-(t)). \end{aligned}$$

This is the Black–Scholes formula. □

Corollary 3.3.1 (Black–Scholes put formula). *Let $P(t)$ denote the time- t price of a European put with strike K and maturity T . Then*

$$P(t) = Ke^{-r(T-t)} N(-d_-(t)) - S(t)N(-d_+(t)).$$

Proof. By put–call parity,

$$C(t) - P(t) = S(t) - Ke^{-r(T-t)}.$$

Substituting the call price obtained above and rearranging yields the formula. □

Remark 3. *The argument above derives the Black–Scholes formula directly from the risk-neutral pricing formula. If one writes $C(t) = c(t, S(t))$, then the same function c also solves the Black–Scholes–Merton PDE with terminal condition*

$$c(T, x) = (x - K)^+.$$

So the PDE approach and the risk-neutral expectation approach are simply two different routes to the same pricing formula.

4 The Fundamental Theorem of Asset Pricing in the finite one-period model

We now state the finite-state, one-period version of the Fundamental Theorem of Asset Pricing. This is the precise mathematical statement behind the idea that absence of arbitrage is equivalent to the existence of a risk-neutral measure.

Throughout this subsection, assume that the state space is finite:

$$\Omega = \{\omega_1, \dots, \omega_n\},$$

and that each state has positive probability under the physical measure:

$$P(\{\omega_j\}) > 0, \quad j = 1, \dots, n.$$

This assumption simply means that every state listed in Ω is genuinely possible.

Definition 4.1 (Convex set). *A set $C \subseteq \mathbb{R}^m$ is called convex if for every $x, y \in C$ and every $\lambda \in [0, 1]$,*

$$\lambda x + (1 - \lambda)y \in C.$$

In words: whenever two points lie in the set, the whole line segment joining them also lies in the set.

Theorem 4.1 (Strict separating hyperplane theorem). *Let $C, D \subseteq \mathbb{R}^m$ be nonempty disjoint convex sets. If C is compact and D is closed, then there exist a vector $y \in \mathbb{R}^m$ and real numbers $\alpha < \beta$ such that*

$$y \cdot d \leq \alpha < \beta \leq y \cdot c \quad \forall d \in D, \forall c \in C.$$

Theorem 4.2 (First FTAP: finite one-period version). *The following are equivalent.*

1. The market admits no arbitrage.
2. There exist numbers $q_1, \dots, q_n > 0$ such that

$$\pi^i = \sum_{j=1}^n q_j S^i(\omega_j), \quad i = 0, 1, \dots, d.$$

3. There exists a probability measure $\mathbb{P}^*$ on Ω with

$$\mathbb{P}^*(\omega_j) > 0, \quad j = 1, \dots, n,$$

such that

$$\pi^i = E^* \left[\frac{S^i}{1+r} \right], \quad i = 0, 1, \dots, d.$$

Definition 4.2 (Completeness). *The market is called complete if for every contingent claim*

$$X : \Omega \rightarrow \mathbb{R}$$

there exists a portfolio $\bar{\xi} \in \mathbb{R}^{d+1}$ such that

$$X(\omega) = \bar{\xi} \cdot \bar{S}(\omega) \quad \forall \omega \in \Omega.$$

In other words, every payoff can be replicated by trading in the available assets.

Proof. We prove

$$(2) \implies (3), \quad (3) \implies (2), \quad (2) \implies (1), \quad (1) \implies (2).$$

$(2) \implies (3)$. Assume there exist $q_1, \dots, q_n > 0$ such that

$$\pi^i = \sum_{j=1}^n q_j S^i(\omega_j), \quad i = 0, 1, \dots, d.$$

Apply this to the bond, for which $\pi^0 = 1$ and $S^0(\omega_j) = 1 + r$ for every j . Then

$$1 = \pi^0 = \sum_{j=1}^n q_j (1 + r) = (1 + r) \sum_{j=1}^n q_j.$$

Hence

$$\sum_{j=1}^n q_j = \frac{1}{1+r}.$$

Now define

$$\mathbb{P}^*(\omega_j) := (1 + r)q_j, \quad j = 1, \dots, n.$$

Since each $q_j > 0$, we have $\mathbb{P}^*(\omega_j) > 0$, and

$$\sum_{j=1}^n \mathbb{P}^*(\omega_j) = (1 + r) \sum_{j=1}^n q_j = 1.$$

So $\mathbb{P}^*$ is a probability measure on Ω . Moreover,

$$\pi^i = \sum_{j=1}^n q_j S^i(\omega_j) = \frac{1}{1+r} \sum_{j=1}^n \mathbb{P}^*(\omega_j) S^i(\omega_j) = E^* \left[\frac{S^i}{1+r} \right].$$

Thus (3) holds.

(3) $\implies$ (2). Now assume there exists a probability measure $\mathbb{P}^*$ such that

$$\pi^i = E^* \left[\frac{S^i}{1+r} \right], \quad i = 0, 1, \dots, d.$$

Define

$$q_j := \frac{\mathbb{P}^*(\omega_j)}{1+r}, \quad j = 1, \dots, n.$$

Then each $q_j > 0$, and

$$\pi^i = \frac{1}{1+r} \sum_{j=1}^n \mathbb{P}^*(\omega_j) S^i(\omega_j) = \sum_{j=1}^n q_j S^i(\omega_j), \quad i = 0, 1, \dots, d.$$

So (2) holds.

(2) $\implies$ (1). Assume there exist strictly positive state prices $q_1, \dots, q_n$ satisfying

$$\pi^i = \sum_{j=1}^n q_j S^i(\omega_j), \quad i = 0, 1, \dots, d.$$

Suppose, for contradiction, that $\bar{\xi}$ is an arbitrage opportunity. Then

$$\bar{\pi} \cdot \bar{\xi} \leq 0, \quad \bar{\xi} \cdot \bar{S}(\omega_j) \geq 0 \quad \forall j,$$

and since each state has positive probability, there exists at least one state ω_k such that

$$\bar{\xi} \cdot \bar{S}(\omega_k) > 0.$$

Using the pricing relation,

$$\begin{aligned} \bar{\pi} \cdot \bar{\xi} &= \sum_{i=0}^d \pi^i \bar{\xi}^i \\ &= \sum_{i=0}^d \bar{\xi}^i \sum_{j=1}^n q_j S^i(\omega_j) \\ &= \sum_{j=1}^n q_j \sum_{i=0}^d \bar{\xi}^i S^i(\omega_j) \\ &= \sum_{j=1}^n q_j \bar{\xi} \cdot \bar{S}(\omega_j). \end{aligned}$$

Every term on the right-hand side is nonnegative, and the $j = k$ term is strictly positive because $q_k > 0$ and

$$\bar{\xi} \cdot \bar{S}(\omega_k) > 0.$$

Hence

$$\bar{\pi} \cdot \bar{\xi} > 0,$$

which contradicts $\bar{\pi} \cdot \bar{\xi} \leq 0$. Therefore no arbitrage can exist.

(1) $\implies$ (2). Assume now that the market admits no arbitrage.

We encode each portfolio $\bar{\xi}$ by the vector

$$T(\bar{\xi}) = (\bar{\xi} \cdot \bar{S}(\omega_1), \dots, \bar{\xi} \cdot \bar{S}(\omega_n), -\bar{\pi} \cdot \bar{\xi}) \in \mathbb{R}^{n+1}.$$

Thus the first n coordinates record the payoffs of the portfolio in the n states, and the last coordinate records minus its initial cost.

Define

$$T : \mathbb{R}^{d+1} \rightarrow \mathbb{R}^{n+1}$$

by the formula above, and let

$$M := T(\mathbb{R}^{d+1}) \subseteq \mathbb{R}^{n+1}.$$

Take any $m_1, m_2 \in M$. Then there exist portfolios $\bar{\xi}_1, \bar{\xi}_2 \in \mathbb{R}^{d+1}$ such that

$$m_1 = T(\bar{\xi}_1), \quad m_2 = T(\bar{\xi}_2).$$

Let $\lambda \in [0, 1]$. Since T is linear,

$$\lambda m_1 + (1 - \lambda)m_2 = \lambda T(\bar{\xi}_1) + (1 - \lambda)T(\bar{\xi}_2) = T(\lambda \bar{\xi}_1 + (1 - \lambda)\bar{\xi}_2).$$

Because $\lambda \bar{\xi}_1 + (1 - \lambda)\bar{\xi}_2 \in \mathbb{R}^{d+1}$, the right-hand side belongs to M . Hence M is convex.

In fact, because T is linear, M is a linear subspace of $\mathbb{R}^{n+1}$. Since we are in finite dimensions, every linear subspace is closed. So M is both closed and convex.

Now define

$$\Delta := \left\{ z = (z_1, \dots, z_{n+1}) \in \mathbb{R}^{n+1} : z_k \geq 0 \ \forall k, \sum_{k=1}^{n+1} z_k = 1 \right\}.$$

This is the standard simplex in $\mathbb{R}^{n+1}$, so it is nonempty, compact, and convex.

We now show that

$$M \cap \Delta = \emptyset.$$

Suppose, to the contrary, that there exists some portfolio $\bar{\xi}$ such that

$$T(\bar{\xi}) \in M \cap \Delta.$$

Write

$$T(\bar{\xi}) = (\bar{\xi} \cdot \bar{S}(\omega_1), \dots, \bar{\xi} \cdot \bar{S}(\omega_n), -\bar{\pi} \cdot \bar{\xi}).$$

Because $T(\bar{\xi}) \in \Delta$, all coordinates are nonnegative and their sum is 1. Therefore

$$\bar{\xi} \cdot \bar{S}(\omega_j) \geq 0 \quad \forall j, \quad \bar{\pi} \cdot \bar{\xi} \leq 0,$$

and at least one coordinate is strictly positive.

There are now two possibilities.

Case 1: one of the first n coordinates is strictly positive. Then for some state ω_k ,

$$\bar{\xi} \cdot \bar{S}(\omega_k) > 0.$$

So the portfolio $\bar{\xi}$ has nonnegative payoff in every state, strictly positive payoff in at least one state, and nonpositive initial cost. Since each state has positive probability, this is an arbitrage opportunity.

Case 2: the first n coordinates are all zero. Then

$$\bar{\xi} \cdot \bar{S}(\omega_j) = 0 \quad \text{for all } j = 1, \dots, n.$$

Since $T(\bar{\xi}) \in \Delta$, its coordinates are nonnegative and sum to 1. Hence the last coordinate must be 1:

$$-\bar{\pi} \cdot \bar{\xi} = 1.$$

Equivalently,

$$\bar{\pi} \cdot \bar{\xi} = -1 < 0.$$

So buying the portfolio $\bar{\xi}$ gives one unit of cash at time 0, while its time-1 payoff is zero in every state.

Now use that one unit of cash to buy one unit of the bond. Define

$$\bar{\eta} := \bar{\xi} + e_0,$$

where $e_0 = (1, 0, \dots, 0)$ denotes one unit of the bond and zero units of all other assets. Then

$$\bar{\pi} \cdot \bar{\eta} = \bar{\pi} \cdot \bar{\xi} + \pi^0 = -1 + 1 = 0.$$

Also, for every state ω_j ,

$$\bar{\eta} \cdot \bar{S}(\omega_j) = \bar{\xi} \cdot \bar{S}(\omega_j) + S^0(\omega_j) = 0 + (1 + r) = 1 + r > 0.$$

Thus $\bar{\eta}$ has zero initial cost and strictly positive payoff in every state, so it is an arbitrage opportunity.

In both cases we obtain a contradiction. Therefore

$$M \cap \Delta = \emptyset.$$

Since M is closed and convex, and Δ is compact and convex, we may apply the strict separating hyperplane theorem. Hence there exist a vector

$$y = (y_1, \dots, y_n, y_{n+1}) \in \mathbb{R}^{n+1}$$

and real numbers $\alpha < \beta$ such that

$$y \cdot m \leq \alpha < \beta \leq y \cdot z \quad \forall m \in M, \forall z \in \Delta.$$

Because $0 \in M$, we have

$$0 = y \cdot 0 \leq \alpha.$$

We now show that in fact

$$y \cdot m = 0 \quad \forall m \in M.$$

Suppose not. Then there exists $m_0 \in M$ such that $y \cdot m_0 \neq 0$.

If $y \cdot m_0 > 0$, then for every $\lambda > 0$ we also have $\lambda m_0 \in M$ because M is a linear subspace. Hence

$$\lambda y \cdot m_0 = y \cdot (\lambda m_0) \leq \alpha,$$

which is impossible for large λ .

If $y \cdot m_0 < 0$, then $-m_0 \in M$ and

$$y \cdot (-m_0) = -y \cdot m_0 > 0,$$

so the previous argument gives a contradiction again.

Thus

$$y \cdot m = 0 \quad \forall m \in M.$$

Next, for each $k = 1, \dots, n+1$, the standard basis vector e_k belongs to Δ . Therefore

$$y_k = y \cdot e_k \geq \beta > 0.$$

So every coordinate of y is strictly positive.

Finally, take any portfolio $\bar{\xi} \in \mathbb{R}^{d+1}$. Since $T(\bar{\xi}) \in M$ and y vanishes on M ,

$$0 = y \cdot T(\bar{\xi}).$$

Expanding this gives

$$\begin{aligned} 0 &= \sum_{j=1}^n y_j (\bar{\xi} \cdot \bar{S}(\omega_j)) - y_{n+1} \bar{\pi} \cdot \bar{\xi} \\ &= \bar{\xi} \cdot \left(\sum_{j=1}^n y_j \bar{S}(\omega_j) - y_{n+1} \bar{\pi} \right). \end{aligned}$$

This holds for every $\bar{\xi} \in \mathbb{R}^{d+1}$. Hence the vector inside the parentheses must be zero

$$\sum_{j=1}^n y_j \bar{S}(\omega_j) - y_{n+1} \bar{\pi} = 0.$$

Equivalently,

$$y_{n+1} \bar{\pi} = \sum_{j=1}^n y_j \bar{S}(\omega_j).$$

Now define

$$q_j := \frac{y_j}{y_{n+1}}, \quad j = 1, \dots, n.$$

Because every coordinate of y is strictly positive, each $q_j > 0$. Dividing the last equation by y_{n+1} gives

$$\bar{\pi} = \sum_{j=1}^n q_j \bar{S}(\omega_j).$$

In components,

$$\pi^i = \sum_{j=1}^n q_j S^i(\omega_j), \quad i = 0, 1, \dots, d.$$

This is (2). □

Theorem 4.3 (Second FTAP: finite one-period version). *Assume the market is arbitrage-free. Then the market is complete if and only if the risk-neutral measure is unique.*

Proof. We prove both implications.

Completeness $\implies$ uniqueness. Assume the market is complete, and let $\mathbb{P}_1^*$ and $\mathbb{P}_2^*$ be two risk-neutral measures.

For each state ω_j , consider the claim

$$X_j(\omega_k) := \mathbf{1}_{\{\omega_j\}}(\omega_k) = \begin{cases} 1, & k = j, \\ 0, & k \neq j. \end{cases}$$

Because the market is complete, there exists a portfolio $\bar{\xi}^{(j)} \in \mathbb{R}^{d+1}$ replicating X_j :

$$X_j(\omega) = \bar{\xi}^{(j)} \cdot \bar{S}(\omega) \quad \forall \omega \in \Omega.$$

Hence the time-0 price of X_j is well defined. Under any risk-neutral measure,

$$\pi(X_j) = \frac{1}{1+r} E^*[X_j].$$

Therefore,

$$\pi(X_j) = \frac{1}{1+r} \mathbb{P}_1^*(\omega_j) \quad \text{and} \quad \pi(X_j) = \frac{1}{1+r} \mathbb{P}_2^*(\omega_j).$$

So

$$\mathbb{P}_1^*(\omega_j) = \mathbb{P}_2^*(\omega_j) \quad \forall j = 1, \dots, n.$$

Since the state space is finite, the two measures agree on every singleton and therefore are identical. Thus the risk-neutral measure is unique.

Uniqueness $\implies$ completeness. Assume now that the risk-neutral measure is unique, but suppose for contradiction that the market is not complete.

Let

$$\mathcal{M} := \left\{ x = (x_1, \dots, x_n) \in \mathbb{R}^n : x_j = \bar{\xi} \cdot \bar{S}(\omega_j) \text{ for some } \bar{\xi} \in \mathbb{R}^{d+1} \right\}$$

be the set of attainable payoffs. Since the market is not complete,

$$\mathcal{M} \neq \mathbb{R}^n.$$

Because $\mathcal{M}$ is a proper linear subspace of $\mathbb{R}^n$, there exists a nonzero vector

$$h = (h_1, \dots, h_n) \in \mathcal{M}^\perp.$$

That is,

$$\sum_{j=1}^n h_j x_j = 0 \quad \forall x = (x_1, \dots, x_n) \in \mathcal{M}.$$

Since the market is arbitrage-free, by the first FTAP there exists at least one risk-neutral measure $\mathbb{P}^*$. Let

$$q_j := \frac{\mathbb{P}^*(\omega_j)}{1+r}, \quad j = 1, \dots, n,$$

be the corresponding state prices. Each q_j is strictly positive.

Choose $\varepsilon \neq 0$ small enough that

$$q_j + \varepsilon h_j > 0 \quad \forall j = 1, \dots, n.$$

This is possible because h is fixed and each $q_j > 0$.

Define

$$q_j^{(\varepsilon)} := q_j + \varepsilon h_j, \quad j = 1, \dots, n.$$

We claim that $(q_j^{(\varepsilon)})$ is another system of state prices.

First, because the bond payoff vector

$$(S^0(\omega_1), \dots, S^0(\omega_n)) = (1+r, \dots, 1+r)$$

belongs to $\mathcal{M}$, and $h \in \mathcal{M}^\perp$, we have

$$0 = h \cdot (1+r, \dots, 1+r) = (1+r) \sum_{j=1}^n h_j.$$

Hence

$$\sum_{j=1}^n h_j = 0.$$

Therefore

$$\sum_{j=1}^n q_j^{(\varepsilon)} = \sum_{j=1}^n q_j + \varepsilon \sum_{j=1}^n h_j = \sum_{j=1}^n q_j = \frac{1}{1+r}.$$

Next fix any asset $i = 0, 1, \dots, d$. Its payoff vector

$$(S^i(\omega_1), \dots, S^i(\omega_n))$$

belongs to $\mathcal{M}$, so again by orthogonality,

$$\sum_{j=1}^n h_j S^i(\omega_j) = 0.$$

Hence

$$\begin{aligned} \sum_{j=1}^n q_j^{(\varepsilon)} S^i(\omega_j) &= \sum_{j=1}^n (q_j + \varepsilon h_j) S^i(\omega_j) \\ &= \sum_{j=1}^n q_j S^i(\omega_j) + \varepsilon \sum_{j=1}^n h_j S^i(\omega_j) \\ &= \sum_{j=1}^n q_j S^i(\omega_j) \\ &= \pi^i. \end{aligned}$$

So $(q_j^{(\varepsilon)})$ also prices every traded asset correctly.

Now define

$$\mathbb{P}_\varepsilon^*(\omega_j) := (1+r)q_j^{(\varepsilon)}, \quad j = 1, \dots, n.$$

Since the $q_j^{(\varepsilon)}$ are strictly positive and sum to $1/(1+r)$, $\mathbb{P}_\varepsilon^*$ is a probability measure, and it is risk-neutral. Because $h \neq 0$ and $\varepsilon \neq 0$, this measure is different from $\mathbb{P}^*$.

This contradicts uniqueness. Therefore the market must be complete. $\square$

4.1 Examples: when risk-neutral measures fail to exist or fail to be unique

Example 1: No risk-neutral measure

Consider a one-period model with one stock and one bond. Let

$$S_0 = 100, \quad B_0 = 1, \quad B_1 = 1.1,$$

and suppose

$$S_1 = \begin{cases} 130, & \text{up state,} \\ 115, & \text{down state.} \end{cases}$$

If a risk-neutral measure Q existed, it would have to satisfy

$$S_0 = \frac{1}{1.1} E^Q[S_1],$$

so

$$E^Q[S_1] = 110.$$

But under any probability measure, the expectation of S_1 must lie between its two possible values 115 and 130. Hence $E^Q[S_1] = 110$ is impossible. Therefore no risk-neutral measure exists.

This reflects the fact that the model admits arbitrage: the stock outperforms the bond in every state.

Example 2: Multiple risk-neutral measures

Now consider a one-period model with one stock and one bond, but with three possible future states. Let

$$S_0 = 100, \quad B_0 = 1, \quad B_1 = 1.05,$$

and suppose

$$S_1 = \begin{cases} 130, & \text{state } u, \\ 105, & \text{state } m, \\ 80, & \text{state } d. \end{cases}$$

A risk-neutral measure Q must satisfy

$$100 = \frac{1}{1.05} E^Q[S_1],$$

that is,

$$E^Q[S_1] = 105.$$

Writing

$$Q(u) = q_u, \quad Q(m) = q_m, \quad Q(d) = q_d,$$

we require

$$q_u + q_m + q_d = 1,$$

and

$$130q_u + 105q_m + 80q_d = 105.$$

Substituting $q_m = 1 - q_u - q_d$, we obtain

$$130q_u + 105(1 - q_u - q_d) + 80q_d = 105,$$

which simplifies to

$$q_u = q_d.$$

Hence the family of risk-neutral measures is

$$q_u = q_d = a, \quad q_m = 1 - 2a,$$

with $0 \leq a \leq \frac{1}{2}$. If we require equivalence to the physical measure, then all probabilities must be strictly positive, so

$$0 < a < \frac{1}{2}.$$

Thus there are infinitely many equivalent martingale measures.

For example, both

$$Q_1 = (0.2, 0.6, 0.2) \quad \text{and} \quad Q_2 = (0.4, 0.2, 0.4)$$

are risk-neutral measures.

This nonuniqueness reflects market incompleteness: with only one stock and one bond but three future states, not every contingent claim can be replicated.

Exercises

1. Let $T > 0$ be fixed and divide the interval $[0, T]$ into n equal subintervals of length

$$\Delta t = \frac{T}{n}.$$

In the Cox-Ross-Rubinstein model, the stock price over one time step moves by a factor

$$u_n = e^{\sigma\sqrt{\Delta t}}, \quad d_n = e^{-\sigma\sqrt{\Delta t}},$$

where $\sigma > 0$ is fixed.

1. Show that

$$u_n d_n = 1.$$

2. Using the Taylor expansion of the exponential function around 0, show that

$$u_n = 1 + \sigma\sqrt{\Delta t} + \frac{1}{2}\sigma^2\Delta t + O((\Delta t)^{3/2}),$$

and

$$d_n = 1 - \sigma\sqrt{\Delta t} + \frac{1}{2}\sigma^2\Delta t + O((\Delta t)^{3/2}).$$

3. Deduce that the one-step relative change in the stock price is of order

$$\sqrt{\Delta t}.$$

4. Explain why a scaling of order $\sqrt{\Delta t}$ is natural if one hopes to obtain a nondegenerate random limit over a fixed time horizon $[0, T]$.
5. Compare this with the alternative scaling

$$u_n = 1 + \sigma\Delta t, \quad d_n = 1 - \sigma\Delta t.$$

What goes wrong with this choice if one wants a diffusion limit?

2. In the CRR model above, assume the bond grows by the factor

$$e^{r\Delta t}$$

over one time step. The one-step risk-neutral probability is therefore

$$p_n^* = \frac{e^{r\Delta t} - d_n}{u_n - d_n}.$$

1. Write $u_n - d_n$ explicitly and use Taylor expansion to show that

$$u_n - d_n = 2\sigma\sqrt{\Delta t} + O((\Delta t)^{3/2}).$$

2. Expand $e^{r\Delta t}$ and d_n up to order Δt and show that

$$e^{r\Delta t} - d_n = \sigma\sqrt{\Delta t} + \left(r - \frac{1}{2}\sigma^2\right)\Delta t + O((\Delta t)^{3/2}).$$

3. Divide the two expansions to prove that

$$p_n^* = \frac{1}{2} + \frac{r - \frac{1}{2}\sigma^2}{2\sigma}\sqrt{\Delta t} + o(\sqrt{\Delta t}).$$

4. Explain why this shows that, for very small time steps, the risk-neutral probability is close to $1/2$, but not exactly equal to $1/2$.
5. What is the role of the correction term

$$\frac{r - \frac{1}{2}\sigma^2}{2\sigma}\sqrt{\Delta t}?$$

Give an intuitive interpretation.

3. Let $Y_k^{(n)}$ denote the one-step log-return in the n -step CRR model:

$$Y_k^{(n)} = \begin{cases} \log u_n, & \text{with probability } p_n^*, \\ \log d_n, & \text{with probability } 1 - p_n^*. \end{cases}$$

Recall that

$$\log u_n = \sigma\sqrt{\Delta t}, \quad \log d_n = -\sigma\sqrt{\Delta t}.$$

1. Show that

$$E^*[Y_k^{(n)}] = p_n^*\sigma\sqrt{\Delta t} + (1 - p_n^*)(-\sigma\sqrt{\Delta t}) = (2p_n^* - 1)\sigma\sqrt{\Delta t}.$$

2. Using the expansion for p_n^* from the previous exercise, deduce that

$$E^*[Y_k^{(n)}] = \left(r - \frac{1}{2}\sigma^2\right)\Delta t + o(\Delta t).$$

3. Show that

$$E^*[(Y_k^{(n)})^2] = \sigma^2\Delta t.$$

4. Use

$$\text{Var}^*(Y_k^{(n)}) = E^*[(Y_k^{(n)})^2] - (E^*[Y_k^{(n)}])^2$$

to prove that

$$\text{Var}^*(Y_k^{(n)}) = \sigma^2\Delta t + o(\Delta t).$$

5. Interpret the two formulas above. Why do they suggest that, under the risk-neutral measure, the log-price should behave approximately like a normal random walk with drift

$$r - \frac{1}{2}\sigma^2$$

and variance rate σ^2 ?

4. Let $S_n(T)$ denote the stock price at time T in the n -step CRR model. Show that

$$\log S_n(T) = \log S_0 + \sum_{k=1}^n Y_k^{(n)}.$$

1. Using the previous exercise, compute the approximate mean of

$$\sum_{k=1}^n Y_k^{(n)}.$$

2. Compute the approximate variance of

$$\sum_{k=1}^n Y_k^{(n)}.$$

3. Explain why the $Y_k^{(n)}$ are independent under the risk-neutral measure.
 4. Using the Central Limit Theorem heuristically, argue that as $n \rightarrow \infty$,

$$\log S_n(T)$$

should converge in distribution to a normal random variable with mean

$$\log S_0 + \left(r - \frac{1}{2}\sigma^2\right) T$$

and variance

$$\sigma^2 T.$$

5. Deduce heuristically that the limiting stock price should satisfy

$$S(T) = S_0 \exp \left(\left(r - \frac{1}{2}\sigma^2\right) T + \sigma Z \right),$$

where $Z \sim N(0, T)$.

6. Explain why this is exactly the terminal distribution of geometric Brownian motion under the risk-neutral measure.

5. Let

$$C_n(0) = e^{-rT} E_n^*[(S_n(T) - K)^+]$$

be the price of a European call option in the n -step CRR model.

1. Explain why this is the correct no-arbitrage price in the n -step binomial model.
 2. Using the previous exercise, argue heuristically that as $n \rightarrow \infty$,

$$S_n(T)$$

should converge in distribution to a lognormal random variable.

3. If one informally passes to the limit inside the pricing formula, what limiting expression does one obtain for the call price?
 4. Show that this limiting expression is

$$C(0) = e^{-rT} E^Q[(S(T) - K)^+],$$

where under Q ,

$$\log S(T) \sim N \left(\log S_0 + \left(r - \frac{1}{2}\sigma^2\right) T, \sigma^2 T \right).$$

5. Explain, in words, why the CRR model can be viewed as a discrete approximation to the Black-Scholes model.

6. Consider a one-period market with one bond and one stock. The bond satisfies

$$B_0 = 1, \quad B_1 = 1.05.$$

The stock satisfies

$$S_0 = 100, \quad S_1 = \begin{cases} 120, & \omega_1, \\ 90, & \omega_2. \end{cases}$$

1. Find the risk-neutral probability $Q(\omega_1) = q$.
 2. Check that $0 < q < 1$.
 3. Use the first FTAP to decide whether the market is arbitrage-free.
 4. Is the market complete? Explain using the second FTAP.
7. Consider the same bond as above, but now suppose

$$S_0 = 100, \quad S_1 = \begin{cases} 130, & \omega_1, \\ 110, & \omega_2. \end{cases}$$

1. Show that no risk-neutral probability can exist.
 2. Use the first FTAP to conclude that the market admits an arbitrage.
 3. Construct an explicit arbitrage portfolio.
8. Consider a one-period market with one bond and one stock. Let

$$B_0 = 1, \quad B_1 = 1.1,$$

and

$$S_0 = 100, \quad S_1 = \begin{cases} 130, & \omega_1, \\ 110, & \omega_2, \\ 80, & \omega_3. \end{cases}$$

1. Find all risk-neutral measures $Q = (q_1, q_2, q_3)$.
 2. Show that there are infinitely many such measures.
 3. Use the first FTAP to decide whether the market is arbitrage-free.
 4. Use the second FTAP to decide whether the market is complete.
 5. Give an example of a contingent claim that cannot be replicated.
9. In the market from the previous exercise, consider the claim

$$X(\omega_1) = 20, \quad X(\omega_2) = 5, \quad X(\omega_3) = 0.$$

1. Compute

$$\frac{1}{1+r} E^Q[X]$$

for a general risk-neutral measure Q .

2. Does the value depend on the choice of Q ?
3. What does this tell you about whether X is attainable?
4. Find the range of arbitrage-free prices for X .

10. Consider a one-period market with one bond and two stocks. Let

$$B_0 = 1, \quad B_1 = 1.1,$$

and let

$$S_0^1 = 100, \quad S_0^2 = 100.$$

At time 1, the two stocks have payoffs

	ω_1	ω_2	ω_3
S_1^1	130	100	80
S_1^2	90	120	80

1. Find the risk-neutral measure $Q = (q_1, q_2, q_3)$.
 2. Check whether all probabilities are strictly positive.
 3. Use the first FTAP to decide whether the market is arbitrage-free.
 4. Use the second FTAP to decide whether the market is complete.
11. In the market from the previous exercise, consider the contingent claim

$$X(\omega_1) = 10, \quad X(\omega_2) = 20, \quad X(\omega_3) = 0.$$

1. Find its unique no-arbitrage price.
 2. Find a portfolio (ξ^0, ξ^1, ξ^2) that replicates X .
 3. Verify directly that the cost of this portfolio equals the risk-neutral price.
12. Let

$$\Omega = \{\omega_1, \omega_2, \omega_3\}$$

and suppose the market has one bond and two risky assets. Let

$$\bar{S}(\omega_j) = (S^0(\omega_j), S^1(\omega_j), S^2(\omega_j)).$$

Define the payoff matrix

$$A = \begin{pmatrix} S^0(\omega_1) & S^1(\omega_1) & S^2(\omega_1) \\ S^0(\omega_2) & S^1(\omega_2) & S^2(\omega_2) \\ S^0(\omega_3) & S^1(\omega_3) & S^2(\omega_3) \end{pmatrix}.$$

1. Explain why a payoff $X = (X(\omega_1), X(\omega_2), X(\omega_3))$ is attainable if and only if

$$X = A\bar{\xi}$$

for some portfolio $\bar{\xi} \in \mathbb{R}^3$.

2. Show that the market is complete if and only if A has rank 3.
 3. More generally, if there are n states, show that the market is complete if and only if the payoff matrix has rank n .
13. Consider a finite one-period market with n states and $d + 1$ assets. Let

$$A = \begin{pmatrix} S^0(\omega_1) & \cdots & S^d(\omega_1) \\ \vdots & & \vdots \\ S^0(\omega_n) & \cdots & S^d(\omega_n) \end{pmatrix}$$

be the payoff matrix. Let $\bar{\pi} = (\pi^0, \dots, \pi^d)$ be the time-zero price vector.

1. Show that a vector of state prices $q = (q_1, \dots, q_n)$ satisfies

$$A^\top q = \bar{\pi}.$$

2. Explain why the first FTAP says that the market is arbitrage-free if and only if this linear system has a strictly positive solution $q > 0$.
 3. Explain why the second FTAP says that, under no-arbitrage, the market is complete if and only if this strictly positive solution is unique.
14. Suppose a finite one-period market is arbitrage-free but not complete. Let

$$\mathcal{M} = \{x \in \mathbb{R}^n : x_j = \bar{\xi} \cdot \bar{S}(\omega_j) \text{ for some portfolio } \bar{\xi}\}$$

be the set of attainable payoffs.

1. Show that $\mathcal{M}$ is a linear subspace of $\mathbb{R}^n$.
2. Explain why incompleteness means $\mathcal{M} \neq \mathbb{R}^n$.
3. Deduce that there exists a nonzero vector $h \in \mathcal{M}^\perp$.
4. Let q be one strictly positive state-price vector. Show that for small enough $\varepsilon \neq 0$,

$$q^\varepsilon = q + \varepsilon h$$

is also strictly positive.

5. Show that q^ε prices every traded asset correctly.
 6. Conclude that the risk-neutral measure is not unique.
15. Suppose a finite one-period market is complete and arbitrage-free. Let Q_1 and Q_2 be two risk-neutral measures.
1. For each state ω_j , define the Arrow-Debreu claim

$$X_j(\omega_k) = 1_{\{j=k\}}.$$

2. Use completeness to explain why X_j is attainable.
3. Show that its price must satisfy

$$\pi(X_j) = \frac{1}{1+r} Q_1(\omega_j) = \frac{1}{1+r} Q_2(\omega_j).$$

4. Deduce that $Q_1 = Q_2$.
16. Consider a finite one-period market. Suppose that there exists a risk-neutral measure Q , but $Q(\omega_k) = 0$ for some state ω_k , while the physical measure satisfies $P(\omega_k) > 0$.
1. Does this satisfy the version of the first FTAP stated in class?
 2. Explain why the theorem requires strictly positive probabilities under the risk-neutral measure.
 3. Try to construct a payoff that is positive only in state ω_k . What price does Q assign to it?
 4. Discuss why this is economically suspicious if ω_k is genuinely possible.
17. Let X be a contingent claim in an arbitrage-free finite one-period market. Suppose the market is incomplete and let $\mathcal{Q}$ denote the set of risk-neutral measures.

1. Show that if X is attainable, then

$$\frac{1}{1+r}E^Q[X]$$

is the same for every $Q \in \mathcal{Q}$.

2. Show that if there exist $Q_1, Q_2 \in \mathcal{Q}$ such that

$$E^{Q_1}[X] \neq E^{Q_2}[X],$$

then X is not attainable.

3. Interpret this result in terms of the second FTAP.

18. Consider a one-period model with one stock, one bond, and three states:

$$B_0 = 1, \quad B_1 = 1 + r,$$

$$S_0 = s_0, \quad S_1(\omega_1) = s_1, \quad S_1(\omega_2) = s_2, \quad S_1(\omega_3) = s_3,$$

where

$$s_1 > s_2 > s_3 > 0.$$

1. Show that a risk-neutral measure exists if and only if

$$s_3 < (1+r)s_0 < s_1.$$

2. Assuming this condition holds, describe the set of all risk-neutral measures.
3. Is the market complete? Explain.
4. What changes if there are only two states instead of three?